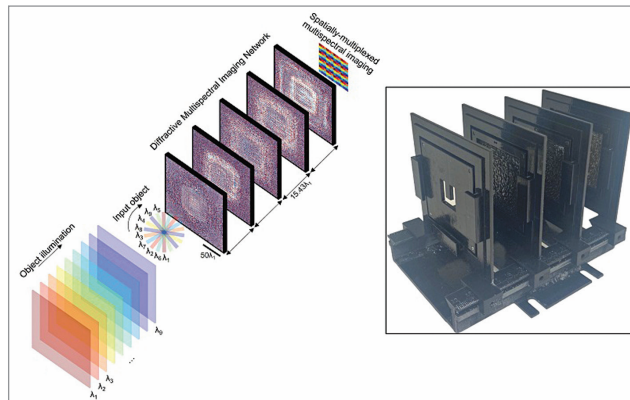


Diffractive optical network facilitates snapshot multispectral quantitative phase imaging

Researchers at the University of California, Los Angeles (UCLA), have showcased a multispectral quantitative phase imaging (QPI) blueprint. This setup taps into the power of deep learning to create a wide-ranging diffractive optical



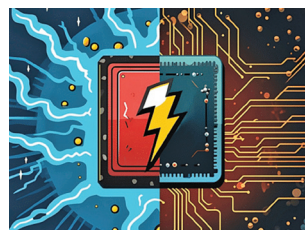
This diffractive multispectral imager can convert a monochrome image sensor into a snapshot multispectral imaging device without conventional spectral filters or digital reconstruction algorithms (Credit: Ozcan Lab, UCLA)

network, which facilitates the capture of quantitative phase images over multiple spectral bands in a singular snapshot. After assembling the diffractive layers, the optical network functions as a comprehensive phase-to-intensity converter, extracting phase information from input subjects at preset wavelengths by directing multispectral QPI signals to specific locations on the output plane. A single-colour focal plane array then measures intensity patterns, with pixel clusters analysing QPI signals linked to distinct spectral bands. Numerical

simulations confirmed the device's effectiveness, demonstrating its capability to perform QPI across 16 spectral bands within the visible light spectrum while maintaining uniform efficiency. Successful imaging trials, including human cell imagery, validated the design's robustness and adaptability.

Light-electron system boosts eco-friendly computing

MIT researchers have tapped into the potential of photonics to boost current computing speeds, showcasing its effectiveness in machine learning. Their innovation, named 'Lightning' utilises a reconfigurable photonic-electronic SmartNIC to aid



MIT researchers introduce Lightning, a reconfigurable photonic-electronic smartNIC that serves real-time deep neural network inference requests at 100Gbps (Credit: Alex Shipp/MIT CSAIL via Midjourney)

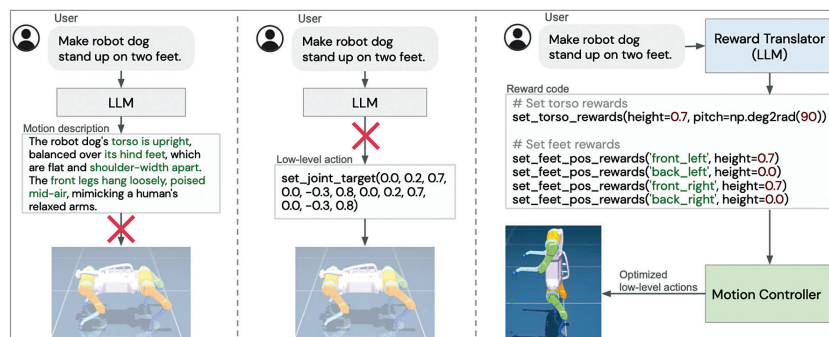
deep neural networks, akin to machine-learning models that simulate the human brain's data processing method, in executing inference tasks such as image recognition and generating responses in chatbots like ChatGPT. The team's hybrid system uses a reconfigurable count-action abstraction, merging photonics and electronics to control data flows. It translates

electron-carried data into photons for quick inference, then switches them back to electrons for communication, ensuring Lightning's rapid real-time computing without previous start-and-stop delays. 'Lightning' technology uses photons for faster, cooler computation. Compared to conventional units like GPUs and SmartNICs, Lightning's chip demonstrated superior energy efficiency in handling inference requests.

Robot skill development through language models

Researchers at Google DeepMind aim to develop an interactive framework that harnesses large language models (LLMs) to master foundational robotic tasks. They devised the reward translator to transform user interactions into robot

motion reward functions. Utilising the open source MJPC tool on the MuJoCo simulator, they established a sequence of fine-tuned actions for each control interval, which the robot then executes in real-time, facilitating the next planning stage



Approach overview (Credit: <https://language-to-reward.github.io>)

through MJPC updates. The system was tested with two simulated robots—a quadruped and a dexterous manipulator, demonstrating its effectiveness through various tasks. Leveraging the F-VLM detector on a mobile manipulator, they identified image objects, filtered background noise in point cloud data, and measured the z-axis from an aerial perspective, effectively showcasing sim-to-real transfer in object handling, with accurate reward code generation and desired motion creation through Mujoco MPC.